SATO: Straighten Any 3D Tubular Object

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***Abstract*—3D tubular objects have complex spatial shapes. Direct volume visualization cannot intuitively display their morphological characteristics and surface abnormalities. Straightening reformation is an effective visualization method for tubular objects. It uses a swept frame to sample cross sections along the centerline of the tubular object to generate straightening result. So far, however, current methods cannot visualize the full 3D view and fail to interface with downstream morphological analysis. Furthermore, current swept frames impose strict restrictions on the shape of tubular objects and are computationally expensive. In this study, we propose a novel swept frame based on vector rotation and construct an automatic straightening reformation pipeline. Our method is applicable to various tubular objects and can be efficiently executed recursively while ensuring that the straightening results have no rotation bias. Extensive experiments on eight different tubular objects and quantitative evaluation on various downstream applications demonstrate the effectiveness and universality of our straightening pipeline. Code is available at https://github.com/Yanfeng-Zhou/SATO.**

***Index Terms*—Straighten, tubular object, swept frame, visualization, curved planar reformation.**

I. Introduction

3D TUBULAR objects often have a slender and curved shape. Direct volume visualization based on 3D images has occlusion problems and cannot reflect complete morphological information, which makes it difficult for some downstream analyses, such as colon polyp identification [1], [2], spinal deformity assessment [3], [4] and vessel wall calcification detection [5], [6].

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In order to more intuitively display the morphological characteristics and surface abnormalities of tubular objects, some related studies have proposed various visualization methods, such as maximum intensity projection, multi-planar reformation, direct volume rendering, and straightening reformation.

Straightening reformation is an effective method. It samples cross sections along the centerline of the tubular object to generate a linear straightening result. The straightening results can be completely displayed in the same view, eliminating occlusion problems and interference from other non-target structures [7].

The common straightening methods are Straightened Curved Planar Reformation (SCPR) and its improvements [5], [6], [8], [9], [10]. They often use Frenet [11] as the swept frame to establish the local coordinate system for each point on the centerline of the tubular object, then use strategies such as interest line projection, ray casting, etc. to resample the cross sections and connect them sequentially, finally project them to 2D view to generate the straightening result.

However, these straightening methods straighten only raw images and cannot interface with automatic morphological analysis. Furthermore, their straightening results are only displayed in 2D view and require multiple adjustments to the visualization angle to describe a relatively complete spatial structure. More importantly, Frenet frame relies on second derivative to establish the local coordinate system, which requires that the second derivative of centerline of the tubular object cannot be 0. This imposes strict restrictions on the shape of tubular object (for example, the tubular object cannot contain line segments and planar segments with inflection points) and cannot be efficiently performed by computational programs.

In this study, we propose a novel straightening reformation pipeline SATO. SATO uses the raw 3D image as input, first segments the tubular object, then extracts centerline based on the segmentation result and marks the starting and ending points, and finally straightens the raw image and the segmentation result based on centerline to generate the corresponding straightened result. For the straightening process, to solve the limitations of Frenet frame, we propose Zhou's swept frame. Our frame is based on vector rotation and represented as an efficient recursive form, which is applicable to various tubular objects without shape restrictions and ensures that the straightening results have no rotation bias. Furthermore, our straightening reformation pipeline can not only directly generate complete 3D views and multi-angle 2D views for visualization, but also perform fully automatic morphological analysis (such as stenosis rate calculation, aneurysm detection, coronary artery lesion analysis, etc.) based on straightened results. We compare SATO with previous straightening methods and conduct extensive experiments on eight different

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tubular objects, including colon, blood vessels, spine, microtubules, mitochondria, neurons, conical helices and double helices. We also demonstrate its practical significance on various downstream tasks.

Our contributions are summarized as follows:

- We propose a fully-automatic straightening reformation pipeline SATO which is applicable to various 3D tubular objects.
- We propose Zhou's swept frame that can be efficiently executed by computational programs. Our frame has no restrictions on the centerlines of tubular objects and ensures that the straightening results have no rotation bias.
- Extensive experiments on multiple downstream tasks and qualitative results on tubular objects with different shapes demonstrate the effectiveness and universality of our method.

II. Related Work

A. Straightened Curved Planar Reformation and its Improvements

Straightened Curved Planar Reformation (SCPR) is one of three common Curved Planar Reformations (the others are Projected CPR and Stretched CPR), which can straighten tubular objects for visualization [8]. It first extracts the centerline of tubular object, then uses Frenet frame to establish the local coordinate system of each cross section with the point on centerline as the origin, and finally resamples the cross sections sequentially and projects them along the predefined interest lines to generate straightening results. Some researchers have improved SCPR for centerline extraction, cross section resampling strategy and swept frame. For centerline extraction, [12] proposes a fast and robust centerline calculation for multi-contrast 3D magnetic resonance imaging data, [13] proposes a 3D thinning algorithm for centerline extraction. For cross section resampling strategy, [1] uses curved and disjoint electric field lines for resampling, which avoids double and inadequate sampling of cross sections at high curvatures. Reference [5] proposes helical interest line projection to generate multi-angle views from the outside to the inside. Because the swept frame is an important component of straightening reformation with our focus and improvement, we further analyze their calculation methods and corresponding characteristics in the following paragraph. We compare and evaluate the characteristics of these straightening methods in Table II of Section IV-B.

B. Swept Frame

In order to straighten the tubular object, we need to establish the corresponding coordinate system for each cross section. They consist of three mutually orthogonal unit vectors $\vec{x}_i$, $\vec{y}_i$ and $\vec{z}_i$ (also called normal vector, binormal vector and tangent vector respectively), where i represents the i-th point on the centerline, i.e. the i-th cross section. The method for determining $\vec{x}_i$, $\vec{y}_i$ and $\vec{z}_i$ is called the swept frame. The common swept frame is Frenet from differential geometry. It uses the second derivative to define the normal vector $\vec{x}_i$:

$$\vec{x}_i = \gamma''(i)/|\gamma''(i)|, \tag{1}$$

where $\gamma(i) = (x(i), y(i), z(i))$ represents the centerline, $x(i)$, $y(i)$ and $z(i)$ represent 3D coordinates. Frenet imposes a strict restriction on curve shape: the second derivative of the centerline cannot be 0 (such as line segments and planar segments with inflection points). Reference [11] proposes a modified Frenet frame for more general curves. It first determines whether $\gamma(i)$ has planar segments and treats planar segments and non-planar segments separately: for the non-planar segment, follow the calculation method of Frenet; for the planar segment, use a fixed vector $\vec{m}$ perpendicular to the plane where the planar segment is located as $\vec{x}_i$. Modified Frenet eliminates the restriction that the second derivative is 0 but also brings rotation bias at the connection point between the planar and non-planar segments. Bishop [14] proposes a rotation minimization frame. It defines $\vec{x}_i$ and $\vec{y}_i$ based on differential equations:

$$\begin{aligned} \vec{x}_i' &= -(\gamma''(i) \cdot \vec{x}_i)\gamma'(i)/|\gamma'(i)|, \\ \vec{y}_i' &= -(\gamma''(i) \cdot \vec{y}_i)\gamma'(i)/|\gamma'(i)|. \end{aligned} \tag{2}$$

It is extremely computationally expensive. In contrast, our frame is based on vector rotation, which has low computational cost and no rotation bias. We compare different swept frames in Table I of Section IV-A.

C. Applications of Straightening Reformation

Straightening reformation has been widely used in colon flattening, vascular abnormality detection, spine visualization, etc. Reference [15] uses SCPR to straighten the spine to facilitate navigation and positioning in 3D space. Reference [16] visualizes tubular flow based on straightening reformation such as arterial blood flow measurement and tubular gas flow simulation from the automotive industry. Reference [17] proposes a vessel refinement method combining SCPR and optimal path finding method. Reference [5] uses Helical CPR for vessel wall calcification detection. Reference [18] uses straightening reformation to analyze the correlation between cerebral artery stenosis and cognitive impairment in elderly. Our straightening pipeline can be used for visualization and morphological analysis. We perform four practical applications (intuitive visualization of complex tubular objects, aneurysm detection, vessel stenosis grade assessment and coronary arteriosclerosis risk prediction) to demonstrate its significance for computer-aided diagnosis in Section IV-D.

III. Method

In Section III-A, we introduce the overall flow of our straightening pipeline, and then elaborate on the specific implementation strategies of each module, including segmentation, centerline extraction and straightening reformation. In order to ensure that the straightening reformation can be applied to all tubular objects and be efficiently executed by computational programs, we propose a fast and efficient swept frame based on vector rotation in Section III-B. Finally, we further elaborate on the complete process of straightening reformation based on our swept frame in Section III-C.

A. Straightening Pipeline for Any 3D Tubular Object

Figure 1 shows the overall flow of straightening pipeline. We use the raw 3D image as input and segments the tubular

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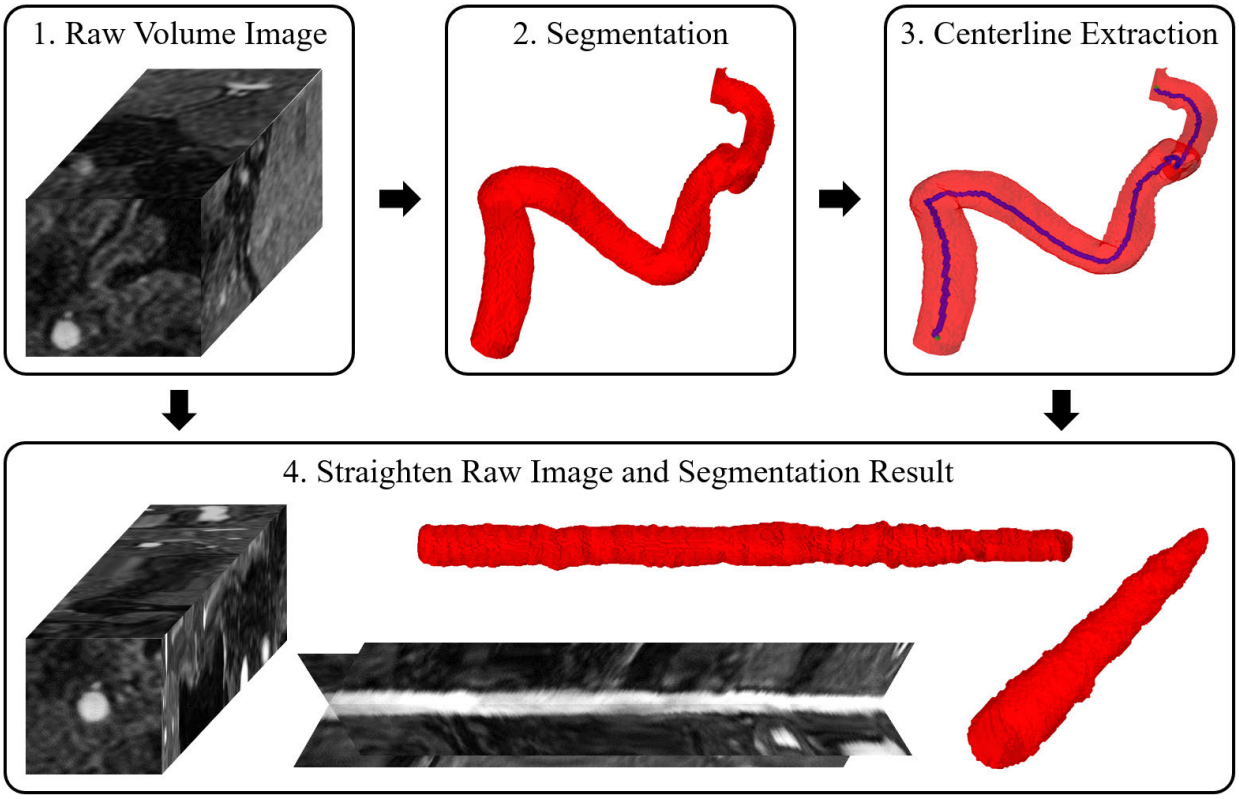

Fig. 1. The overall flow of SATO. Step 4 shows the corresponding straightening results of the raw image and segmentation result under different views.

object. Then we extract the centerline based on the segmentation result, and finally use a swept frame to establish the local coordinate system along the centerline to straighten the raw image and corresponding segmentation result.

1) Volumetric Image Segmentation: Volumetric image segmentation has achieved remarkable success with the development of deep neural networks (DNNs). Some studies propose efficient encoder-decoder architectures and achieve superior performance, such as VNet [19], 3D UNet [20], XNet [21], etc. Furthermore, tubular objects commonly manifest a slender morphology. In order to prevent topological errors such as breakage and adhesion during segmentation, researchers have proposed DNNs-based topology-preserving methods, such as clDice [22], Topoloss [23], etc. For different tubular object segmentation, such as colon, mitochondria, vessels, spine, etc., we use the above models comprehensively to generate accurate segmentation results.

2) Centerline Extraction: In order to generate accurate centerline representation, we use a 3D parallel thinning algorithm [13] and take the binary segmentation results as input to extract the centerline. Then we mark the starting, ending and bifurcation points of the centerline. Ideally, the centerline should only have one starting point and one ending point and no bifurcation points. However, due to the interference of abnormal structures and the roughness of segmentation contour, such as burrs, depressions, etc., it may lead to some wrong centerline branches. We need to remove burr branches based on branch length and bifurcation points and finally acquire a complete and correct centerline.

3) Straightening Reformation: We represent the 3D raw image as a cuboid determined by the raw coordinate system. Each voxel is located by a 3D coordinate. We take the raw image and the corresponding centerline as input and use a swept frame to establish the local coordinate system of each cross section with the corresponding point on the centerline as the origin. Then we convert the local coordinates of each point on each cross section into the raw coordinate system and use interpolation to acquire the corresponding voxel intensity. Finally, we stack the resampled cross sections sequentially to reconstruct the straightened result of raw image. Furthermore, as shown in Figure 1, we also generate cutaway illustrations from different angles to better visualize the internal and surface details of tubular objects. The straightening of segmentation result follows the same method.

B. Swept Frame Based on Vector Rotation

We need a swept frame that is applicable to various centerline shapes and efficiently executed by computational programs to address the limitations of previous frames. As shown in Figure 2, consider the $(i\text{-}1)$-th and i-th adjacent points o_{i-1} and o_i on the centerline $\gamma(i)$ and their corresponding cross sections p_{i-1} and p_i, we establish coordinate systems $e_{i-1} = (\vec{x}_{i-1}, \vec{y}_{i-1}, \vec{z}_{i-1})$ and $e_i = (\vec{x}_i, \vec{y}_i, \vec{z}_i)$ with o_{i-1} and o_i as origins, respectively. We use L to represent the length of centerline, then $\gamma(i) = (x(i), y(i), z(i))$, $i \in [0, L]$. We take the unit tangent vectors of the centerline at o_{i-1} and o_i as $\vec{z}_{i-1}$ and $\vec{z}_i$. According to the above definition, we can get:

$$\begin{aligned} \vec{z}_{i-1} &= \gamma'(i-1)/|\gamma'(i-1)|, \\ \vec{z}_{i-1} &\perp p_{i-1}, \\ \vec{z}_i &= \gamma'(i)/|\gamma'(i)|, \\ \vec{z}_i &\perp p_i. \end{aligned} \tag{3}$$

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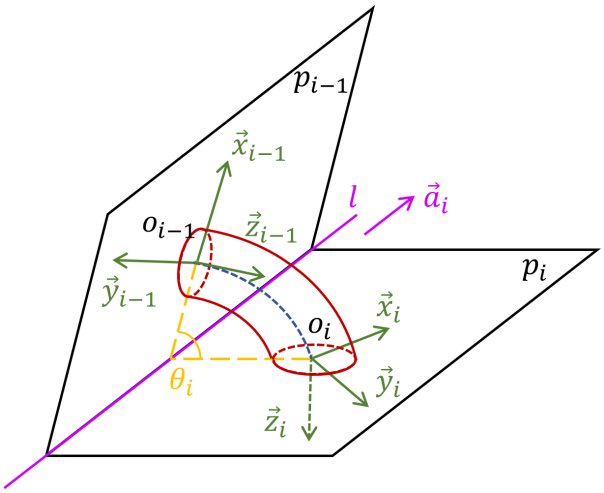

Fig. 2. The coordinate system of the (i-1)-th point o_{i-1} and the i-th point o_i on the centerline. The blue dotted line represents the centerline $\gamma(i)$, the red solid line and dashed line represent the contour of tubular object, the green solid line and dashed line represent the coordinate axes, the purple solid line represents the intersection line l of cross sections p_{i-1} and p_i and the corresponding unit vector $\vec{a}_i$, the yellow dashed line represents the angle between p_{i-1} and p_i, also the angle between $\vec{z}_{i-1}$ and $\vec{z}_i$.

In order to ensure that there is no rotation bias between the two planes after straightening, p_i can be regarded as being acquired by rotating p_{i-1} around the intersection line l of the two planes by an angle θ_i (Note that in practice, the distance between o_{i-1} and o_i is very close, the angle θ_i between the two planes approaches 0, i.e. $\theta_i \approx 0°$). So $\vec{x}_i$ ($\vec{y}_i$) can be considered as acquired by rotating from $\vec{x}_{i-1}$ ($\vec{y}_{i-1}$) around l by an angle θ_i. We use Rodrigues' rotation formula (RRF) to calculate $\vec{x}_i$ ($\vec{y}_i$) from $\vec{x}_{i-1}$ ($\vec{y}_{i-1}$):

$$\begin{aligned}\vec{x}_i &= RRF(\vec{x}_{i-1}, \theta_i, \vec{a}_i) = \vec{x}_{i-1}\cos\theta_i \\ &\quad + (\vec{a}_i \cdot \vec{x}_{i-1}) \cdot \vec{a}_i(1-\cos\theta_i) + (\vec{a}_i \times \vec{x}_{i-1})\sin\theta_i, \\ \vec{y}_i &= RRF(\vec{y}_{i-1}, \theta_i, \vec{a}_i) = \vec{y}_{i-1}\cos\theta_i \\ &\quad + (\vec{a}_i \cdot \vec{y}_{i-1}) \cdot \vec{a}_i(1-\cos\theta_i) + (\vec{a}_i \times \vec{y}_{i-1})\sin\theta_i, \end{aligned} \tag{4}$$

where $\vec{a}_i$ represents the unit direction vector of l, θ_i represents the angle between two planes, i.e. the angle between $\vec{z}_{i-1}$ and $\vec{z}_i$. $\vec{a}_i$ and θ_i are defined as:

$$\begin{aligned}\vec{a}_i &= (\vec{z}_{i-1} \times \vec{z}_i)/|\vec{z}_{i-1} \times \vec{z}_i|, \\ \theta_i &= \cos^{-1}(\vec{z}_{i-1} \cdot \vec{z}_i/(|\vec{z}_{i-1}| \cdot |\vec{z}_i|)). \end{aligned} \tag{5}$$

We have defined $e_i = (\vec{x}_i, \vec{y}_i, \vec{z}_i)$ as a recursive expression, so we only need to define the coordinate system $e_0 = (\vec{x}_0, \vec{y}_0, \vec{z}_0)$ at the starting point of centerline to recursively calculate the complete swept frame. Since $\vec{z}_0 = \gamma'(0) = (x'(0), y'(0), z'(0))^T$, we might as well set:

$$\begin{aligned}\vec{x}_0 &= (0, -z'(0), y'(0))^T, \\ \vec{y}_0 &= \vec{z}_0 \times \vec{x}_0, \end{aligned} \tag{6}$$

and normalize $\vec{z}_0$, $\vec{x}_0$ and $\vec{y}_0$ to unit vectors:

$$\begin{aligned}\vec{z}_0 &= \vec{z}_0/|\vec{z}_0|, \\ \vec{x}_0 &= \vec{x}_0/|\vec{x}_0|, \\ \vec{y}_0 &= \vec{y}_0/|\vec{y}_0|. \end{aligned} \tag{7}$$

At this point, we have constructed a complete swept frame. In our frame, $\vec{z}_i$ is defined by the unit tangent vector of the centerline, $\vec{x}_i$ and $\vec{y}_i$ are acquired by rotating $\vec{x}_{i-1}$ and $\vec{y}_{i-1}$, respectively, which ensures the orthogonality of $\vec{x}_i$, $\vec{y}_i$ and $\vec{z}_i$. We take $\vec{x}_i$ and $\vec{z}_i$ as examples and prove their orthogonality in Proof 1.

Proof 1 $\vec{x}_i \perp \vec{z}_i$

To prove $\vec{x}_i \perp \vec{z}_i$, **just prove** $\vec{x}_i \cdot \vec{z}_i = 0$

$$\begin{aligned}\vec{x}_i \cdot \vec{z}_i &= (\vec{x}_{i-1}\cos\theta_i + (\vec{a}_i \cdot \vec{x}_{i-1}) \cdot \vec{a}_i(1-\cos\theta_i) \\ &\quad + (\vec{a}_i \times \vec{x}_{i-1})\sin\theta_i) \cdot \vec{z}_i\end{aligned}$$

Because $\vec{a}_i \perp \vec{z}_i$, $\vec{a}_i \cdot \vec{z}_i = 0$, **so**

$$\begin{aligned}\vec{x}_i \cdot \vec{z}_i &= \vec{x}_{i-1} \cdot \vec{z}_i \cos\theta_i + (\vec{a}_i \times \vec{x}_{i-1}) \cdot \vec{z}_i \sin\theta_i \\ &= \vec{x}_{i-1} \cdot \vec{z}_i \cos\theta_i + \vec{a}_i \cdot (\vec{x}_{i-1} \times \vec{z}_i)\sin\theta_i\end{aligned}$$

Substitute $\vec{a}_i = (\vec{z}_{i-1} \times \vec{z}_i)/|\vec{z}_{i-1} \times \vec{z}_i|$

$$\begin{aligned}\vec{x}_i \cdot \vec{z}_i &= \vec{x}_{i-1} \cdot \vec{z}_i \cos\theta_i + \\ &\quad (\vec{z}_{i-1} \times \vec{z}_i) \cdot (\vec{x}_{i-1} \times \vec{z}_i)\sin\theta_i/|\vec{z}_{i-1} \times \vec{z}_i| \\ &= \vec{x}_{i-1} \cdot \vec{z}_i \cos\theta_i + \\ &\quad (\vec{z}_{i-1} \times \vec{z}_i) \cdot (\vec{x}_{i-1} \times \vec{z}_i)/|\vec{z}_{i-1}||\vec{z}_i|\end{aligned}$$

Based on Lagrange's identity, we get

$$\begin{aligned}(\vec{z}_{i-1} \times \vec{z}_i) \cdot (\vec{x}_{i-1} \times \vec{z}_i) &= (\vec{z}_{i-1} \cdot \vec{x}_{i-1})(\vec{z}_i \cdot \vec{z}_i) \\ &\quad -(\vec{z}_{i-1} \cdot \vec{z}_i)(\vec{z}_i \cdot \vec{x}_{i-1})\end{aligned}$$

Assume $\vec{z}_{i-1} \perp \vec{x}_{i-1}$, $\vec{z}_{i-1} \cdot \vec{x}_{i-1} = 0$, **then**

$$(\vec{z}_{i-1} \times \vec{z}_i) \cdot (\vec{x}_{i-1} \times \vec{z}_i) = -(\vec{z}_{i-1} \cdot \vec{z}_i)(\vec{z}_i \cdot \vec{x}_{i-1})$$

$$\begin{aligned}\vec{x}_i \cdot \vec{z}_i &= \vec{x}_{i-1} \cdot \vec{z}_i \cos\theta_i - (\vec{z}_{i-1} \cdot \vec{z}_i)(\vec{z}_i \cdot \vec{x}_{i-1})/|\vec{z}_{i-1}||\vec{z}_i| \\ &= \vec{x}_{i-1} \cdot \vec{z}_i \cos\theta_i - \vec{z}_i \cdot \vec{x}_{i-1}\cos\theta_i \\ &= 0\end{aligned}$$

So $\vec{x}_i \perp \vec{z}_i$

For assumption $\vec{z}_{i-1} \perp \vec{x}_{i-1}$, **because** $\vec{x}_0 \perp \vec{z}_0$

So $\vec{z}_{i-1} \perp \vec{x}_{i-1}$ **naturally holds** □

C. Complete Process of Straightening Reformation

Algorithm 1 shows the complete process of straightening reformation. For a given raw image (segmentation result) of the tubular object and its corresponding centerline $\gamma(i)$, we use a cubic spline to fit $\gamma(i)$ to make it smooth and differentiable. Calculate the coordinate system $e_i = (\vec{x}_i, \vec{y}_i, \vec{z}_i)$ of any point o_i on $\gamma(i)$ based on our swept frame. Calculate the rotation matrix R_i and translation matrix M_i between the local coordinate system e_i and the raw coordinate system e:

$$\begin{aligned}R_i &= e^T \cdot e_i, \\ M_i &= o_i^T. \end{aligned} \tag{8}$$

where $e = (\vec{e}_x, \vec{e}_y, \vec{e}_z)$, generally $\vec{e}_x = (1, 0, 0)^T$, $\vec{e}_y = (0, 1, 0)^T$, $\vec{e}_z = (0, 0, 1)^T$. Construct the transformation matrix T_i based on R_i and M_i:

$$T_i = \begin{bmatrix} R_i & M_i \\ \mathbf{O} & \mathbf{1} \end{bmatrix} \tag{9}$$

Construct the local coordinate matrix C_i^{local} of the cross section p_i in the local coordinate system e_i:

$$C_i^{local} = C(W)* = \begin{bmatrix} -W + n//(2W+1) \\ -W + n\%(2W+1) \\ 0 \\ 1 \end{bmatrix}$$
$$(n = 0, 1, \ldots, (2W+1)^2), \tag{10}$$

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W represents the cross section width, which is used to determine the size of the straightened cross section. The value of W is related to the cross sectional radius of tubular object and the size of visual view. Smaller W leads to lower computational costs, but we need to ensure that W is greater than the maximum radius among all cross sections. The symbols // and % represent the quotient and remainder, respectively.

Algorithm 1 Straightening Reformation

Input: Raw Image or Segmentation Result,
Centerline $\gamma(i) = (x(i), y(i), z(i)), i \in [0, L]$
Parameter: Raw Coordinate System $e = (\vec{e}_x, \vec{e}_y, \vec{e}_z)$,
Cross Section Width W
Output: Straightening Result F

Cubic spline fitting $\gamma(i)$
for $i = 0$ **to** L **do**
  $\vec{z}_i \leftarrow (x'(i), y'(i), z'(i))^T$
  $\vec{z}_i \leftarrow \vec{z}_i / |\vec{z}_i|$
  if $i = 0$ **then**
    $\vec{x}_i \leftarrow (0, -z'(0), y'(0))^T$
    $\vec{y}_i \leftarrow \vec{z}_i \times \vec{x}_i$
    $\vec{x}_i \leftarrow \vec{x}_i / |\vec{x}_i|$
    $\vec{y}_i \leftarrow \vec{y}_i / |\vec{y}_i|$
  else
    $\vec{a}_i \leftarrow \vec{z}_{i-1} \times \vec{z}_i / |\vec{z}_{i-1} \times \vec{z}_i|$
    $\theta_i \leftarrow \cos^{-1}(\vec{z}_{i-1} \cdot \vec{z}_i / (|\vec{z}_{i-1}| \cdot |\vec{z}_i|))$
    $\vec{x}_i \leftarrow RRF(\vec{x}_{i-1}, \theta_i, \vec{a}_i)$
    $\vec{y}_i \leftarrow RRF(\vec{y}_{i-1}, \theta_i, \vec{a}_i)$
  end if
  $e_i \leftarrow (\vec{x}_i, \vec{y}_i, \vec{z}_i)$
  $R_i \leftarrow e^T \cdot e_i$
  $M_i \leftarrow \gamma(i)^T$
  $T_i \leftarrow \begin{bmatrix} R_i & M_i \\ \mathbf{O} & \mathbf{1} \end{bmatrix}$
  $C_i^{local} \leftarrow C(W)$
  $C_i^{raw} \leftarrow T_i C_i^{local}$
  $f_i \leftarrow I(C_i^{raw})$
end for
$F \leftarrow [f_0, f_1, \ldots, f_L]$
Return: F

According to the transformation matrix T_i, calculate raw coordinate matrix C_i^{raw} of the cross section p_i in raw coordinate system e:

$$C_i^{raw} = T_i C_i^{local}. \tag{11}$$

According to the raw coordinate matrix C_i^{raw}, use image interpolation to calculate the pixel intensity (segmentation category) matrix f_i of the cross section p_i:

$$f_i = I(C_i^{old}), \tag{12}$$

where $I(\cdot)$ represents the interpolation method. Bilinear interpolation is often used for volumetric images, while nearest neighbor interpolation is often used for segmentation results. Stack the cross sections together sequentially to generate final straightening result F:

$$F = [f_0, f_1, \ldots, f_L]. \tag{13}$$

IV. Comparison, Demonstration and Application

A. Comparison of Swept Frame

We compare our frame with previous swept frame, including Frenet [11], modified Frenet [11] and Bishop [14]. Table I shows their calculation method, time complexity, strengths and weaknesses. Other frames either have restrictions on curve shapes, or treat planar and non-planar segments separately, while our frame is applicable for various 3D curves and has no rotation bias between adjacent points during straightening reformation. Furthermore, compared with calculation methods based on the second derivative or differential equation, our frame only requires vector rotation, can be efficiently executed recursively and achieves the lowest time complexity.

B. Comparison of Straightening Methods

We compare SATO with previous straightening methods, including SCPR [8], Angelelli et al. [16], Nonlinear Colon Unfolding (NCU) [2], Multi-Angle SCPR [6], Saroul et al. [30], Curve-Centric Volume Reformation (CCVR) [31], Helical CPR [5], Electrical Field Line (EFL) [1], Volumetric SCPR [9], MIP-CPR [10].

Table II shows their swept frames, cross section resampling strategies and visualization views. For curve sweeping, our frame has greater universality and higher efficiency. For cross section resampling, the common strategies include interest line projection, ray casting, 2D planes and surfaces. Interest line projection can produce 2D views along interest lines with different shapes, such as straight lines, helical lines, and multi-angle lines. Ray casting samples by tracing rays emitted from the center point to the surface of tubular object, which results in higher sampling efficiency but lower accuracy. The 2D plane can visualize the complete view of tubular object and surrounding space, but it requires sampling all coordinate points of the cross section, resulting in lower sampling efficiency. Surfaces (such as Coons surfaces, electric field lines, etc.) can effectively avoid the artifacts caused by cross section intersection at high centerline curvature during the reformation process. However, it comes with the lowest sampling efficiency. For visualization view, our method can simultaneously straighten the raw images and the corresponding segmentation results and generate 3D views and multi-angle 2D cutaway illustrations.

Furthermore, we evaluate different methods in terms of automaticity, interactivity with downstream analysis and open-source. Our method is fully automatic and can flexibly interface with downstream morphological analysis. We also provide open-source code for utilization and extension.

C. Qualitative Demonstration of Straightening Results

Figure 3 shows the straightening results of different tubular objects, including colon, blood vessels, conical helices, double helices, spine, microtubules, mitochondria and neurons. We can see that tubular objects can be effectively straightened regardless of their curvature, cross sectional shape, length and thickness. Taking spine and mitochondria as examples, we show the straightening results of raw images at different angles in Figure 4. We can see that the straightened raw images can completely display the shape of tubular objects under different views. Furthermore, we can also observe the

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TABLE I
COMPARISON OF DIFFERENT SWEPT FRAMES

Swept Frame	Calculation Method	Definition Formula	Time Complexity	Strength ↑ and Weakness ↓
Frenet	Second Derivative	$\begin{cases} \vec{z}_i = \gamma'(i)/\lvert\gamma'(i)\rvert \\ \vec{x}_i = \gamma''(i)/\lvert\gamma''(i)\rvert \\ \vec{y}_i = \vec{z}_i \times \vec{x}_i \end{cases}$	O(L^2)	↓ Strict restrictions on curve shape: second derivative cannot be 0 (such as line segments and planar segments with inflection points) ↓ The calculation of $\vec{x}_i$, $\vec{y}_i$ requires $\gamma(i)$ is arc-length parameterized, but in practice $\gamma(i)$ is often represented by piecewise polynomials, which results in more complicated expressions for $\vec{x}_i$, $\vec{y}_i$
modified Frenet	Second Derivative	Non-planar segment: consistent with Frenet Planar segment: $\begin{cases} \vec{z}_i = \gamma'(i)/\lvert\gamma'(i)\rvert \\ \vec{x}_i = \vec{m} \text{ (fixed vector)} \\ \vec{y}_i = \vec{x}_i \times \vec{z}_i \end{cases}$	O(L^2)	↑ Applicable to various curves, no restrictions on curve shape ↓ Need to determine whether $\gamma(i)$ has planar segments and treat planar and non-planar segments separately ↓ For straightening reformation, it has rotation bias at the connection of planar and non-planar segments
Bishop	Differential Equation	$\begin{cases} \vec{z}_i = \gamma'(i)/\lvert\gamma'(i)\rvert \\ \vec{x}'_i = -(\gamma''(i)\cdot\vec{x}_i)\gamma'(i)/\lvert\gamma'(i)\rvert \\ \vec{y}'_i = -(\gamma''(i)\cdot\vec{y}_i)\gamma'(i)/\lvert\gamma'(i)\rvert \end{cases}$	O(L^2)~O(L^3)	↓ $\gamma(i)$ is required to consist of one or more second-order differentiable segments ↓ Need to solve a complex differential equation for each point i
Zhou (Ours)	Vector Rotation	$\begin{cases} \vec{z}_i = \gamma'(i)/\lvert\gamma'(i)\rvert \\ \vec{x}_i = RRF(\vec{x}_{i-1}, \theta_i, \vec{a}_i) \\ \vec{y}_i = RRF(\vec{y}_{i-1}, \theta_i, \vec{a}_i) \\ \vec{a}_i = \vec{z}_{i-1} \times \vec{z}_i/\lvert\vec{z}_{i-1} \times \vec{z}_i\rvert \\ \theta_i = \cos^{-1}(z_{i-1}\cdot z_i/(\lvert z_{i-1}\rvert\cdot\lvert z_i\rvert)) \end{cases}$	O(L)	↑ Applicable to various curves, no restrictions on curve shape ↑ Represent sweeping as an efficient recursive process ↑ Rotation angles of three coordinate axes between adjacent points are the same, without rotation bias for straightening

TABLE II
COMPARISON OF DIFFERENT STRAIGHTENING REFORMATION. - INDICATES THAT THE FRAME USED IS NOT MENTIONED. AUTOMATICITY INDICATES WHETHER IT IS FULLY AUTOMATIC. INTERFACE INDICATES WHETHER IT CAN INTERFACE WITH DOWNSTREAM ANALYSIS. OPEN-SOURCE INDICATES WHETHER THE IMPLEMENTATION IS OPEN-SOURCE

Method	Component			Evaluation		
	Swept Frame	Cross Section	Visualization View	Automaticity	Interface	Open-Source
SCPR	Frenet	Interest Line	Single 2D	✓		✓
Angelelli et al.	modified Frenet	2D Plane	Side-by-Side 2D			
NCU	-	Nonlinear Ray Casting + Nonlinear Scaling	Single 2D	✓		
Multi-Angle SCPR	Frenet	Multi-Angle Interest Line	Multi-Angle 2D	✓		
Saroul et al.	-	Coons Surface	Single 2D			✓
CCVR	modified Frenet	2D Plane or Radial Ray Casting	Multi-Angle 2D and 3D			
Helical CPR	Frenet	Helical Interest Line	Multi-Angle 2D	✓		
SCPR w/ EFL	Frenet	Electric Field Line	Single 2D			
Volumetric SCPR	Frenet	Ray Casting + Interest Line	Single 2D			
MIP-CPR	Bishop	2D plane	Single 2D and 3D	✓		
SATO (Ours)	Zhou	2D Plane	Multi-Angle 2D and 3D	✓	✓	✓

interior and surface of tubular objects from different angles, which effectively solves problems in direct visualization, such as occlusion and failure to display the interior.

D. Practical Applications of SATO

In order to further demonstrate the practical significance of SATO and its ability to automatically interface with downstream analysis, we apply SATO to four practical scenarios, including intuitive visualization of complex tubular objects to avoid self- and extrinsic occlusion, aneurysm detection, stenosis rate calculation of blood vessels and representation learning for coronary arteriosclerosis risk prediction.

1) Intuitive Visualization Avoiding Occlusion: 3D tubular objects suffer from self- and extrinsic occlusions, which prevent direct observation of some details from any viewing angle. As shown in Figure 5 (a), the vascular clusters of mice brain have many branches with complex connections. Occlusion between different vessels prevents direct observation. In contrast, SATO can mark each vessel based on centerline coordinates and perform intuitive straightening visualization. The cracks and protrusions of helical structure in Figure 5 (b) are obscured by itself, but we can accurately locate these cracks and protrusions in straightening results. Based on SATO, we can also extract and straighten nine cerebral arteries from the cerebral artery segmentation results and the process is shown in Figure 5 (c).

2) Aneurysm Detection: As shown in Figure 6 (a), aneurysms often manifest as abnormal bulges and enlargements of vessels. Based on the straightening results, we can determine whether there are continuous cross sections with abnormal radii to detect aneurysms. First we calculate the radius r_i of the i-th cross section based on the cross sectional area A_i:

$$r_i = \sqrt{\frac{A_i}{\pi}}. \tag{14}$$

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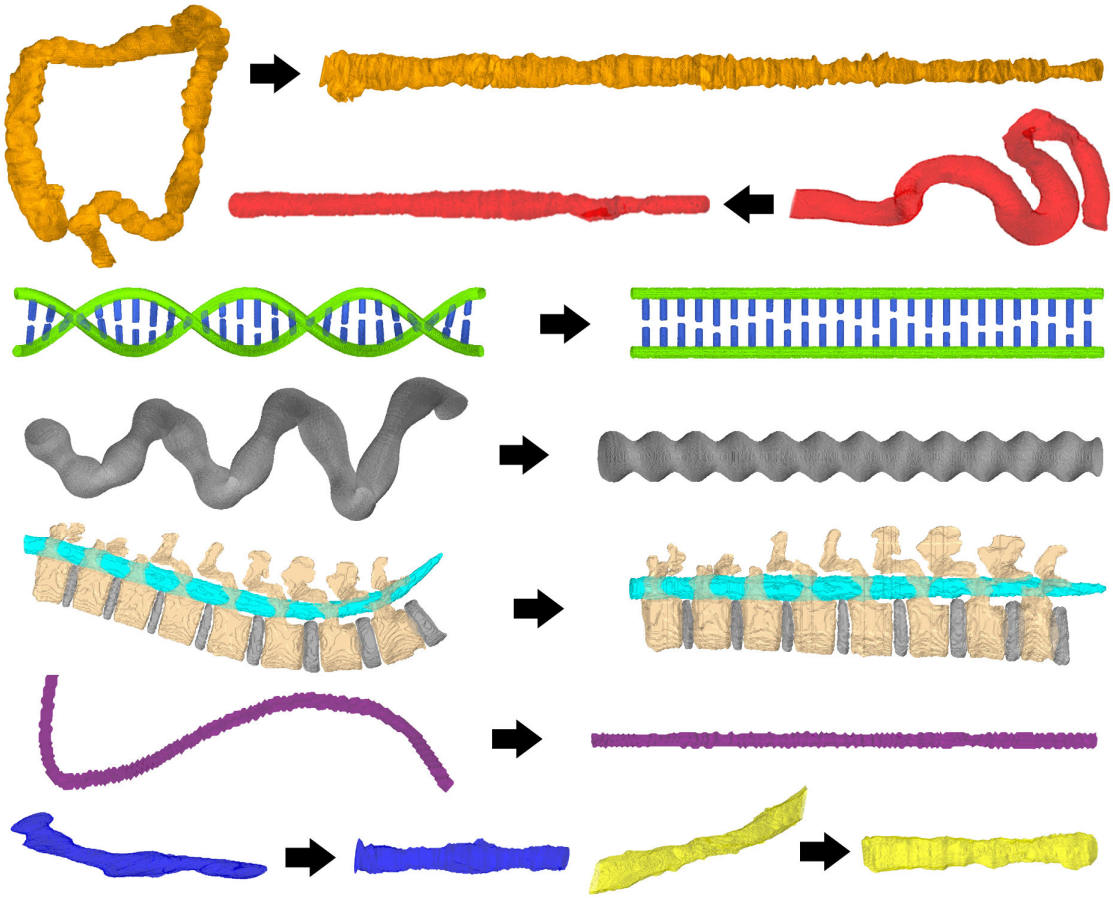

Fig. 3. Qualitative show of straightening results for different tubular objects. From top to bottom: colon (from WORD [24]), blood vessel (from IntrA [25]), synthetic double helix, synthetic conical helix, spine (from SPIDER [26]), microtubule (from BetaSeg [27]), mitochondria (from MitoEM [28]), neuron (from XPRESS [29]). To facilitate visualization of the 3D morphology of tubular objects, we only show the segmentation-based straightening results.

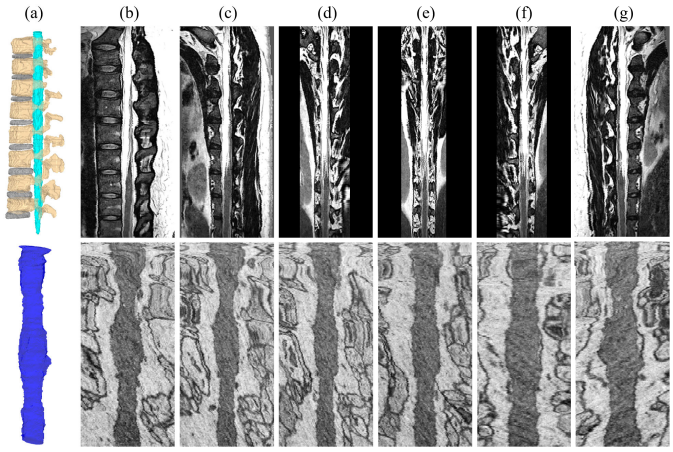

Fig. 4. Taking spine and mitochondria as examples, qualitative comparison of multi-angle 2D views of straightened raw images. (a) straightened segmentation results. (b), (c), (d), (e), (f) and (g) represent the cutaway illustrations from 0°, 30°, 60°, 90°, 120° and 150°, respectively.

Then calculate the mean radius $\bar{r}$ of the entire vessel:

$$\bar{r} = \frac{r_0 + r_1 + \ldots + r_L}{L}, \tag{15}$$

where L represents the number of cross sections, i.e. the length of the tubular object.

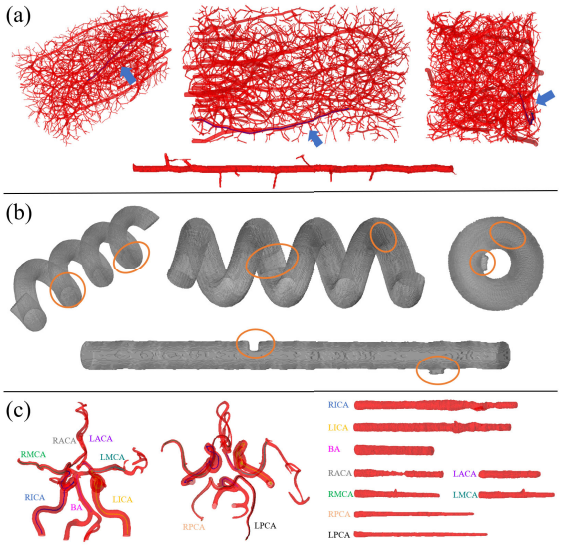

Fig. 5. Some practical scenarios for straightening visualization. (a) Individual vessel visualization in vascular clusters of mice brain. (b) Location of cracks and protrusions in helical structure. (c) Straightening visualization of nine cerebral arteries, including right internal carotid artery (RICA), left internal carotid artery (LICA), basilar artery (BA), right anterior cerebral artery (RACA), left anterior cerebral artery (LACA), right middle cerebral artery (RMCA), left middle cerebral artery (LMCA), right posterior cerebral artery (RPCA) and left posterior cerebral artery (LPCA).

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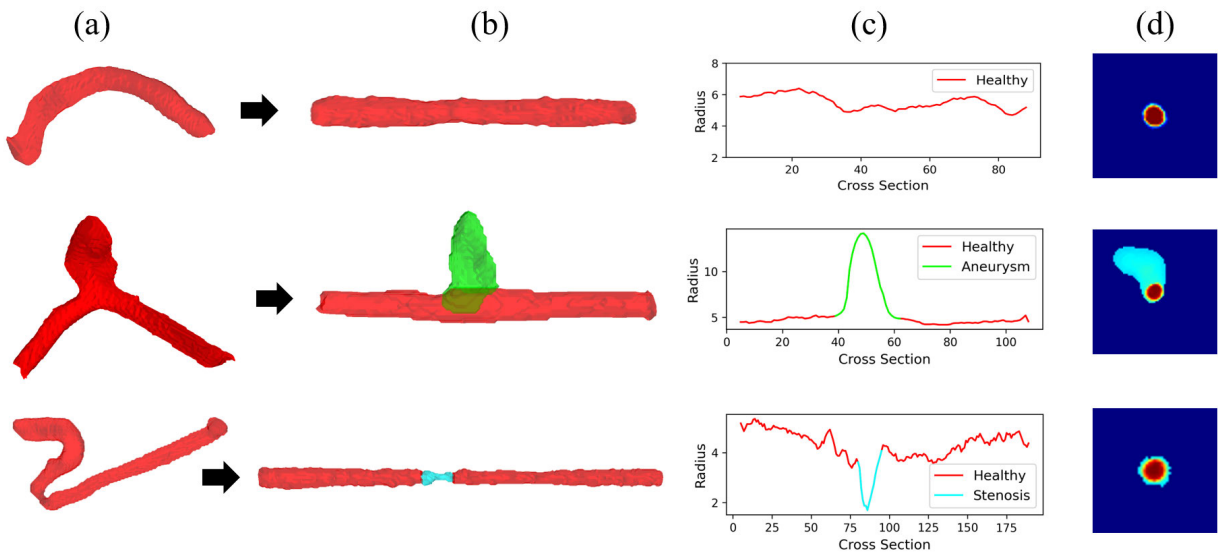

Fig. 6. Visualization of healthy, aneurysmal and stenotic vessels. For top to bottom: healthy vessel, vessel with an aneurysm, stenotic vessel. (a) Raw vessel segments. (b) Straightened vessel segments. Green represents aneurysm and cyan represents stenotic segment. (c) Cross sectional radii. (d) Cross sectional projected heatmap.

TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT METHODS IN ANEURYSM DETECTION AND STENOSIS GRADING

Method	Aneurysm			Mild Stenosis			Moderate Stenosis			Severe Stenosis		
	F1	Sen	Pre	F1	Sen	Pre	F1	Sen	Pre	F1	Sen	Pre
Distance Mapping	68.22	63.77	73.33	**91.41**	**89.85**	93.04	82.28	**82.53**	82.04	68.28	**78.47**	60.43
Inscribed Sphere	67.61	69.57	65.75	87.53	81.37	94.70	79.38	77.31	81.57	67.76	71.53	64.38
Cylindrical Intensity Model	75.91	75.36	76.47	90.94	86.76	95.53	78.25	73.69	83.41	64.38	68.06	63.51
STAO (Ours)	**80.28**	**82.61**	**78.08**	91.08	86.20	**96.55**	**82.35**	78.71	**86.34**	**70.55**	75.69	**66.06**

Finally, we detect vessel segments with abnormal radii and locate the aneurysm based on the difference $(r_i - \bar{r})$ between the radius of each cross section and the mean radius $\bar{r}$. Figure 6 (b) and (c) show the straightened blood vessel shape and the radius of each cross section, respectively. The cross sectional projected heatmaps in Figure 6 (d) illustrates the abnormal shape.

To evaluate the performance of aneurysm detection, we collect 108 cerebral magnetic resonance angiography images from different subjects and annotate aneurysms by two radiologists with more than five years of experience. We segment blood vessels, extract corresponding centerlines and topologies. Then we locate and extract the cerebral arteries and finally acquire 611 vessel segments (69 with aneurysms and 542 healthy segments).

We compare SATO with other radius-aware methods, including distance mapping, inscribed sphere and cylindrical intensity model. Table III illustrates the comparison results. Our model achieves better sensitivity (Sen), precision (Pre) and F1-score (F1). Note that distance mapping and inscribed sphere have worse results. It may be because distance mapping only computes local minimum radii along the centerline and fails to detect some saccular aneurysms. Inscribed sphere is also insensitive to some small saccular aneurysms.

3) Stenosis Rate Calculation: Vascular stenosis often manifests as a dramatic decrease in cross sectional area and radius. We can calculate the stenosis rate s based on the mean cross sectional radius $\bar{r}$ and the minimum cross sectional radius r_{min} of the straightened vessel:

$$s = \frac{\bar{r} - r_{min}}{\bar{r}} \times 100\%. \tag{16}$$

According to the stenosis ratio s, we can classify the stenosis into three levels: mild ($s < 50\%$), moderate ($50\% \leq s < 75\%$) and severe ($s \geq 75\%$).

In order to evaluate the performance of stenosis grading, we construct a cerebral artery dataset using the same collection process as the aneurysm dataset. It consists of 2070 cerebral arteries, including 1428 mild, 498 moderate and 144 severe

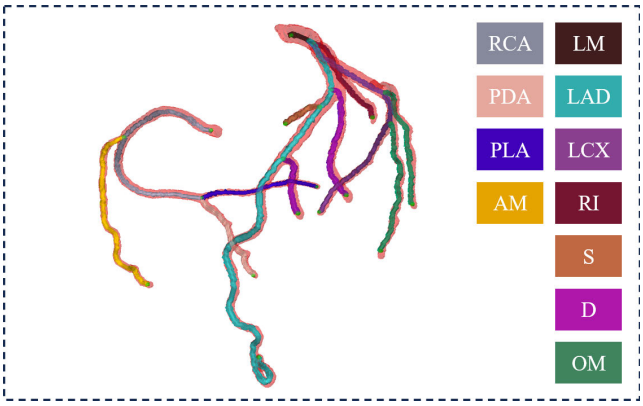

Fig. 7. The components of coronary arteries. Each color indicates a specific branch.

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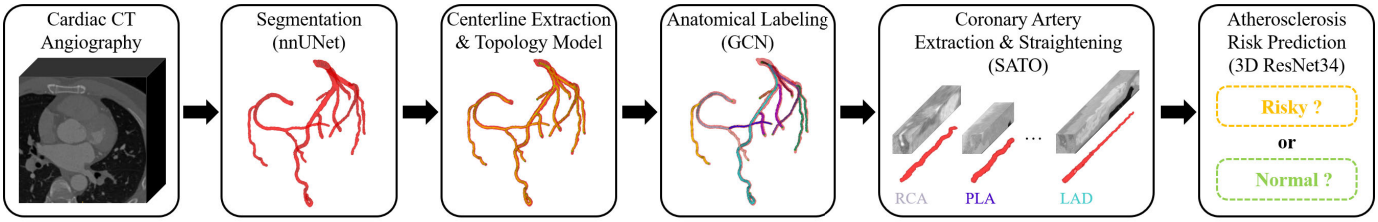

Fig. 8. The overall flow of arteriosclerosis risk prediction. Its core is to use our straightening method STAO to generate straightened coronary artery images containing PCAT and plaque information. Furthermore, segmentation, anatomical labeling and risk prediction are all based on DNNs.

stenoses. Table III compares the performance of various methods. We can see that cylindrical intensity model performs poorly. Cylindrical intensity model calculates radii while segmenting. However, the segmentation may break at stenosis, which has a negative impact on stenosis rate calculation. We also note that distance mapping achieves higher sensitivity but lower precision. It may be because, as mentioned before, distance mapping emphasizes local minimum radii. Overall, our method achieves superior performance, because our method can resample cross-sections and calculate radii based on cross-sectional areas, which makes the calculation more accurate.

4) Coronary Arteriosclerosis Risk Prediction: As shown in Figure 7, the coronary artery consists of left domain (LD) and right domain (RD). LD includes left main (LM), left descending artery (LAD), left circumflex artery (LCX), ramus intermedius (RI), diagonal artery (D), septal artery (S) and obtuse margin (OM). While LAD, LCX and RI are from LM, S and D are from LAD, OM is from LCX. RD includes right coronary artery (RCA), posterolateral artery (PLA), posterior descending artery (PDA) and acute marginal artery (AM). While PLA, PDA and AM are from RCA. Generally, LM, LAD, LCX and RCA are the main branches and the others are side branches.

Some related studies have shown that coronary arteriosclerosis is associated with pericoronary adipose tissue (PCAT) and plaque accumulation [32], [33], [34]. To explore the impact of related factors and predict atherosclerosis risk, we used SATO to construct a computer-aided diagnosis pipeline. The overall flow is shown in Figure 8. Our pipeline takes cardiac CT angiography (CCTA) images as input and uses nnUNet [35] to segment coronary arteries. Then we extract centerlines and construct the corresponding topologies. We use a graph convolutional network (GCN) [36] for anatomical annotation of coronary arteries, and then use SATO to extract and straighten each coronary artery. Finally, we input the straightened images and segmentation results of coronary arteries into 3D ResNet34 [37] to predict arteriosclerosis risk.

To evaluate the risk prediction ability of our pipeline, we collected 511 CCTA images from different subjects and annotated each branch by two radiologists with more than three years of experience. We randomly and equally divide the dataset into five subsets and use five-fold cross-validation to train and evaluate all images. The prediction performance of each coronary artery is shown in Table IV. Overall, our pipeline achieves 79.86%±3.85% for F1-score, 79.53%±4.27% for sensitivity and 80.22%±3.57% precision. We also compare the performance of cropping each coronary artery from the raw image and directly inputting it into 3D ResNet34 [37] in Table IV. This strategy does not perform straightening, so it is affected by irrelevant tissues and cannot focus on the coronary artery and PCAT. Its performance is inferior. This comparison strongly proves the positive effect of straightening on the prediction of arteriosclerosis risk.

TABLE IV
THE PREDICTION PERFORMANCE OF ATHEROSCLEROSIS RISK IN EACH CORONARY ARTERY. W/O STRAIGHTENING INDICATES THAT THE CROPPED CORONARY ARTERIES ARE NOT STRAIGHTENED AND DIRECTLY FED INTO 3D RESNET34. W/ STRAIGHTENING INDICATES THAT USING THE STRAIGHTENED CORONARY ARTERIES AS INPUT

		w/o straightening			w/ straightening		
Branch	Artery	F1	Sen	Pre	F1	Sen	Pre
Main	RCA	77.91	77.29	78.53	82.52	81.46	83.60
	LM	80.53	81.23	79.84	81.71	82.92	80.53
	LAD	75.75	76.49	75.02	83.93	83.10	84.78
	LCX	77.37	77.83	76.92	85.79	86.55	85.05
	RI	76.20	75.05	77.38	78.01	76.56	79.51
Side	PLA	73.26	72.19	74.37	74.27	73.26	75.30
	PDA	71.95	71.09	72.84	76.54	75.58	77.52
	AM	79.32	80.03	78.62	84.32	84.47	84.17
	D	75.28	75.84	74.72	76.94	77.18	76.70
	S	71.81	71.43	72.19	74.68	74.11	75.26
	OM	77.06	76.73	77.40	79.81	79.59	80.03
	Avg	76.04	75.93	76.17	79.86	79.53	80.22
	Std	2.70	3.16	2.38	3.85	4.27	3.57

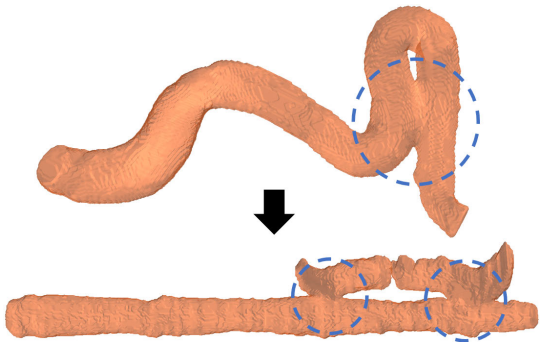

Fig. 9. Qualitative show of straightening distortion. The blue circles highlight the areas of adhesion.

V. LIMITATION

A. Lack of Quantitative Evaluation Metrics For Straightening Quality

Straightening can be considered as a special image enhancement. However, compared to other vision tasks, straightening reformation is a relatively immature and unexplored topic.

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The complete straightening quality evaluation metrics have not been established, because the ground truth of straightening results is not available. It is also our future research goal.

B. Sensitive to Topological Errors of Segmentation

Straightening based on segmentation results is sensitive to topological errors such as breaks and adhesions. Figure 9 shows a tubular object with adhesion in the segmentation result. The straightening result shows severe distortion in the adhesion area.

C. Fail to Take Advantage of Large-Scale Data

We use a swept frame-based spatial transformation to generate straightening results. But this method is not data-driven, i.e., it does not require training with labeled data but cannot take advantage of large-scale data.

VI. Conclusion

We propose an automatic straightening reformation pipeline and a corresponding vector rotation-based swept frame. Our method is applicable to various tubular objects and can be efficiently executed recursively. Extensive experiments on various tubular objects and quantitative evaluation on downstream applications demonstrate the effectiveness and universality of our method.

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